

# CariSurg MedTech Pathways Research Project

## Preliminary Proposal

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**Project Title:** A Bias-Audited ML Triage Tool for Nurse-Led Emergency Care in a Caribbean Regional ED

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### Statement of the Problem

Triage at Mercer General ED relies heavily on paper forms that preclude systematic auditing; comparable regional settings omit critical clinical documentation in up to 67% of low-acuity cases, driving diagnostic error (1). Furthermore, existing artificial intelligence (AI) tools have been shown to underestimate illness severity in patients of colour due to demographic measurement bias (2).

### Previous Work

Overcrowding poses a critical challenge in emergency departments (EDs) due to high patient volumes, restricted resource availability, and the intense cognitive burden of manually prioritising patients under acute time constraints. To address these operational pressures, Ziad and Adda (3) evaluated several machine learning (ML) models—such as Support Vector Machines (SVM) and Gradient Boosting Machines (GBM)—alongside an advanced stacking ensemble model designed to automate patient sorting using standard clinical guidelines. While the individual models displayed strong predictive capacity, the integrated stacking model achieved the highest performance, yielding an accuracy of 80.05% and a precision of 80.27%. However, a significant constraint of their study is its reliance on a singular dataset, which restricts the model's proven adaptability outside of its initial training environment.

In traditional hospital settings, triage systems rely heavily on a nurse's subjective clinical judgement, which introduces operational inconsistencies and bottlenecks during peak traffic periods or major casualty surges (4). Rather than proposing a primary empirical framework, the researchers conducted a comprehensive narrative review of peer-reviewed clinical and engineering literature spanning 2015 to 2024 to synthesise the current landscape of automated prioritisation. Their findings indicate that while AI integrations reduce patient wait times and optimise clinical asset allocation, widespread implementation remains severely obstructed by systemic obstacles. Specifically, the technology is limited by poor underlying hospital data quality, complex ethical concerns regarding algorithmic fairness, and a lack of comprehensive, standardised regulations to ensure algorithms operate safely without introducing clinical harm or demographic bias.

This relentless rise in emergency department volumes has significantly contributed to widespread healthcare professional burnout and compromised triage precision. To evaluate this trajectory, a recent scoping review comprehensively analysed 1,142 distinct citations from 2013 to 2023, systematically isolating 29 high-quality primary research studies tracking real-world patient outcomes (5). The synthesised evidence explicitly confirmed that machine learning models consistently outperform conventional manual protocols by enhancing disease detection speed, predicting hospital admission rates, forecasting length of inpatient stays, and reducing immediate staff cognitive burdens. Despite these operational advantages, the review highlighted critical vulnerabilities, noting that these algorithms demonstrate a tendency toward overconfident misclassifications that clinical staff may accept uncritically. Furthermore, the findings remain geographically isolated, as the investigated articles were exclusively published in United States journals.

In addition to algorithmic limitations, missing data within electronic health records can unintentionally engender racial bias in AI emergency triage tools (6). Through experimental simulation, the researchers demonstrated that reconstructing missing patient history significantly boosts AI predictive accuracy across patient cohorts. However, reducing this data missingness

affected demographic groups unevenly, causing a substantially larger increase in the false positive rate for White patients (5.18%) than for Black patients (2.79%). This disparity occurred partly because Black patients at the studied facility were statistically more likely to leave the emergency room without being seen (15.1% versus 12.6%), meaning their clinical data was never captured by the system. Ultimately, the authors limited their findings to a single hospital environment, comparing only two demographic groups, concluding that AI trained on racially stratified health records risks entrenching existing systemic inequalities.

Compounding these data disparities, an evaluation of how racial bias is measured and mitigated in clinical machine learning models revealed that 67% of analysed studies explicitly confirmed the presence of racial bias within their algorithms (7). While every study that attempted to mitigate this bias was statistically successful—primarily by cleaning and balancing patient data before model training—the investigation uncovered a distinct lack of methodological consistency, with research teams utilising fragmented mathematical formulas to define "fairness". The authors noted that the medical AI field is severely limited by a lack of shared datasets, poor model reporting standards, and a systemic publication bias wherein researchers omit models that fail bias evaluations. Consequently, because the adoption of algorithmic fairness remains in its infancy, the medical community urgently requires standardised reporting rules and open-access data to prevent these tools from propagating healthcare inequities.

## **Proposed Solution**

I propose a machine learning-augmented Emergency Severity Index (ESI) triage tool combining natural language processing (NLP) for chief complaint analysis with a structured classifier. This tool is designed to support, rather than replace, nurse-led ESI decisions. The project will evaluate whether this approach reduces demographic-based misclassification by 20% or more compared to standard ESI protocols alone—thereby improving acuity accuracy to support patient flow and alleviate ED overcrowding—and will be validated against Mercer General ED patient cases.

## References

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